

Streaming 3DGS Virtual Worlds in 6DoF over Next-Generation Networks

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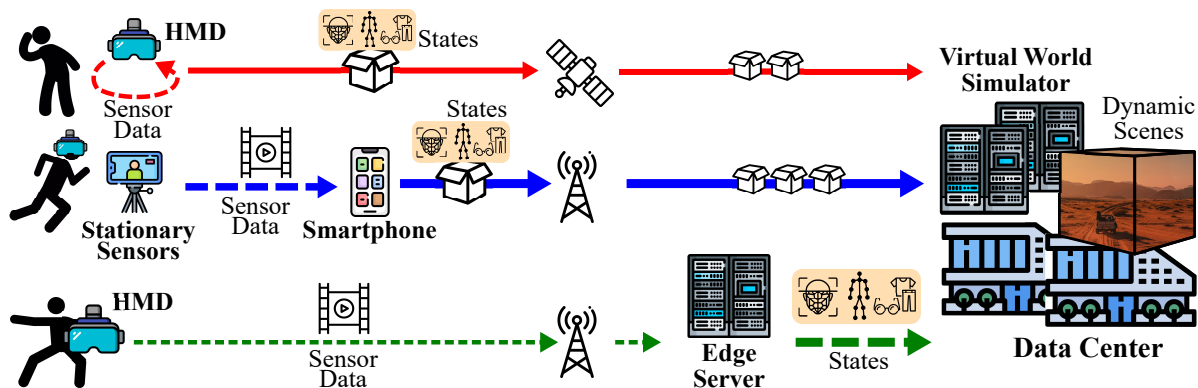


Figure 1: Overview of a 3DGS-based immersive communication system supporting heterogeneous client devices. We only show one-way streaming for clarity, while the reverse direction is symmetric.

Abstract

With the continuous development of networking and computing devices, immersive communication has become increasingly viable, enabling users to explore virtual worlds and interact with other users in 6 Degrees-of-Freedom (6DoF). Immersive communication has great potential not only in professional domains, such as medical diagnostics and distance education but also for leisure activities, such as social networks and new media. This proposal aims to develop an *immersive communication system leveraging the cutting-edge dynamic 3D Gaussians Splatting (3DGS)*. Our objective is to design, implement, and evaluate a highly interactive, adaptive, and efficient immersive communication system that maximizes user experience and system performance while supporting heterogeneous hardware platforms, networks, and applications. We identify and tackle three critical challenges in developing such a system: (i) designing **deformable 3DGS avatars** to enhance real-time user representation, (ii) developing **scalable dynamic 3DGS codecs** to optimize data transmission and storage efficiency, and (iii) implementing **adaptive streaming algorithms** to ensure smooth and responsive user experience across diverse usage scenarios. By solving these challenges, our research aims to push the boundaries of real-time 3D streaming and interaction while

redefining the future of virtual world over next-generation networks.

CCS Concepts

• **Information systems** → **Multimedia streaming**; • **Computing methodologies** → **Mixed / augmented reality**; **Image compression**; • **Networks** → **Data path algorithms**.

Keywords

Volumetric video, streaming, multi-layer representation, adaptive bitrate algorithm

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1 Introduction

Advancements in network technologies such as 5G and 6G cellular networks [7], coupled with increasing computational power in mobile devices and Head-Mounted Displays (HMDs), offer the possibility of streaming rich media data for 6 Degrees-of-Freedom (6DoF) interactions in Virtual Reality (VR), Augmented Reality (AR), and eXtended Reality (XR) applications. These developments have driven the evolution of 3D reconstructions, representations, and rendering technologies, gradually turning immersive communications into reality [6, 17, 33, 35]. Beyond entertainment [17], immersive communications have also been used in teleconferencing [12, 14, 24, 34], telemedicine [3, 25, 39], autonomous systems [13, 20, 23], and other business sectors. While the networking,

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computing, and display technologies keep improving, Jevons Paradox [1] suggests the demand from users for even higher-quality immersive communication experiences will continue to rise, resulting in many more challenges in streaming virtual worlds.

3D Gaussian Splatting (3DGS) [15] has emerged as a compelling choice of 3D representation due to its superior rendering realism, efficiency, and scalability [9]. 3DGS enhances point clouds by adding Gaussian attributes like opacity and transformations as well as Spherical Harmonics (SH) coefficients for view-dependent colors [29]. Compared to 3DGS, 3D meshes [2, 26, 40], 3D point clouds [27], and Neural Radiance Fields (NeRF) [22] suffer from high construction overhead, low visual quality, and slow rendering speed, respectively.

In this proposal, we aim to build a 3DGS-based immersive communication system, as illustrated in Fig. 1, among globally distributed users interacting in a virtual world hosted in data centers. Before joining the virtual world, users first capture and construct their physical likeness into deformable 3DGS avatars. During each communication session, users' states, including body poses, facial expressions, and personal accessories (like glasses and hats), need to be estimated from the sensor data. State estimation can be done with diverse sensors, including mobile sensors on HMDs and smartphones; and stationary sensors like RGB-D cameras and mmWave radars deployed in physical spaces, like meeting rooms and auditorium stages. Sensor data are analyzed by Machine Learning (ML) algorithms on various devices, including HMD devices, smartphones, and edge servers. With estimated states, users' avatars are deformed for other users in the same communication session. Then, the estimated states and deformed avatars are exchanged over the Internet. The rendering task can be performed locally on HMDs or offloaded to nearby devices such as smartphones, personal computers, or edge servers. Upon the states and avatars being sent to the data center, they are placed in a pre-recorded dynamic 3DGS scene within a virtual world simulator. The resulting virtual world is then broadcast to all users to enable real-time interactions among them.

Our considered immersive communication system is general in the sense that heterogeneous clients, diverse rendering devices, and various communication networks can be supported. Fig. 1 shows three sample client devices offloading with decreasing capability (top-down), which rely on mobile sensors, stationary sensors, and ML algorithms on edge servers for estimating user states, respectively. Certain mechanisms need to be developed to realize the immersive communication system, including dynamic 3DGS avatars and scenes. Moreover, several trade-offs must be carefully exercised to ensure user experience with limited resources. For example, performing state estimation on an edge server using cutting-edge ML algorithms may reduce battery consumption of HMDs, but streaming live camera feeds over legacy, say 3G cellular network would definitely lead to excessive delay.

To enable and optimize immersive communications, this proposal focuses on three fundamental research problems:

- (1) **Deformable 3DGS avatars** for realistic and real-time representations of users' states to allow natural interactions.
- (2) **Scalable dynamic 3DGS codecs** for compressing and streaming dynamic 3DGS scenes over diverse and changing networks.

- (3) **Adaptive streaming algorithms** for optimizing the visual quality, interactivity levels, and resource consumption given application-specific requirements.

In the next three sections, we present these research problems along with preliminary research plans.



Figure 2: Human avatars and accessories from our dynamic point cloud dataset [27].



Figure 3: Sample rendered views without (baseline) and with our proposed ABR algorithm (our solution) [29].

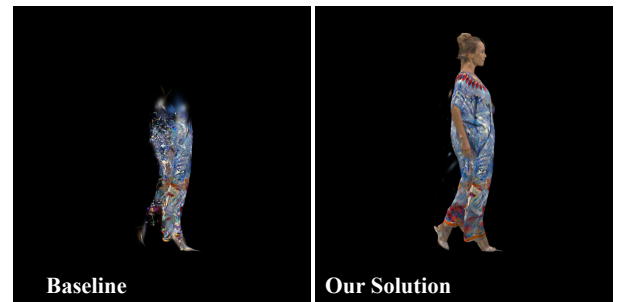


Figure 4: Sample rendered views without (baseline) and with our proposed streaming system (our solution) [28].

2 Deformable 3DGS Avatars

2.1 Proposed Research Tasks

Deformable 3DGS avatars [19, 37] were built offline with default user states, and are deformed online with the latest user states. Essentially their Gaussians are moved/changed to mimic user states,

including body poses and facial expressions. However, existing deformable 3DGS avatars were not developed for real-time and networked applications, leading to three research tasks:

- **Task 1.1: Deformable avatar construction.** Taking photos to construct 3DGS deformable avatar is a time-consuming and tedious process, which requires careful planning. Our prior study [36] indicates the number of input photos and camera positions/orientations affect the visual quality of the constructed 3DGS objects. We plan to develop an algorithm to incrementally compute the next best camera position/orientation using uncertainty metrics of a partially constructed deformable avatar under given resource constraints. Considering the rendered quality of possible deformed avatars during avatar construction is an unexplored problem.
- **Task 1.2: User states estimation.** Detecting user states in real-time from incomplete sensor data remains a challenging problem due to hardware availability or privacy concerns. We plan to develop a scalable ML model that takes mobile and stationary sensors, including inertial and vision sensors, as inputs and produces user states. The scalable ML model can still work with incomplete sensor data, e.g., determining user poses with only front-facing HMD cameras or facial expressions with only stationary RGB-D cameras.
- **Task 1.3: Real-time avatar updates.** Since deformable avatars of all users are exchanged before an immersive communication session starts, only compact user states need to be exchanged. However, the visual quality of deformed avatars under certain states may be affected by the change of physical likeness of user and accessories. We plan to develop a continuous interactive training algorithm that: (i) predicts the rendered quality of deformed avatars, (ii) request mobile/stationary sensors for selective rich media data, and (iii) refine the deformable avatars.

2.2 Prior Work

We have extensive experience with 3D contents, including Multi-View Coding (MVC) [30–32], 3D point cloud [11, 27], and 3DGS [36]. For example, for MVC, we developed camera placement algorithms to enhance synthesized view quality [30, 31]. We also develop a multi-user MVC-based blind streaming system that not only preserves user privacy but also enhances viewing experiences [32]. We extended our success on MVC to 3DGS for an uncertainty-driven path planning algorithm of drones to capture input photos of large scenes [36]. This algorithm can be the starting point of our Task 1.1. Moreover, for 3D point clouds, we built a dynamic dataset [27], including human avatars and accessories with diverse activities as shown in Fig. 2. This figure reveals that complex 3D shapes, such as helmets and clothing, are included in our dataset. Compared to error-prone real experiments, employing our dataset in high-quality renderers, like Unity or Blender, provides ground-truth data for our Tasks 1.1–1.3. These prior studies [27, 30–32, 36] demonstrate our competence in developing deformable 3DGS avatars.

3 Scalable Dynamic 3DGS Codecs

3.1 Proposed Research Tasks

3DGS compression studies can be categorized into two categories: generative and traditional compression [38]. Generative compression focuses on creating more compact 3DGS representations during the reconstruction, whereas traditional compression aims to develop *codecs* to compress trained 3DGS contents into a binary stream to better adapt to devices or network heterogeneity. Existing 3DGS codecs [10, 38] were designed for static 3DGS scenes, which are vulnerable to flickering effects due to randomness in 3DGS training process [15]. To fill up the gap, we identify the following three research tasks:

- **Task 2.1: Hierarchical dynamic representation.** To better accommodate to system and network dynamics, we plan to develop a general dynamic 3DGS representation, which is hierarchical in both spatial and temporal domains. With this new representation, details can be intelligently skipped to minimize the rendered quality degradation while meeting the resource constraints.
- **Task 2.2: Dynamic coding tools.** We plan to develop a set of coding tools to leverage both the spatial and temporal redundancy of dynamic 3DGS scenes. Both traditional signal processing and emerging neural techniques can be evaluated, augmented, and optimized before being added to this tool set.
- **Task 2.3: Hybrid scalable codecs.** We plan to integrate the hierarchical representation with some coding tools into hybrid scalable codecs for potentially-large dynamic 3DGS scenes. Various design alternatives would be proposed and compared for the best-performing design, which would be the first of its kind.

3.2 Prior Work

As a starting point, we released tools for capturing and constructing dynamic 3DGS scenes along with a dataset [42]. Besides, we also released datasets and tools for capturing and analyzing the user behavior in a virtual world [5]. Moreover, we developed a traditional codec for 3DGS scenes by enhancing point cloud codecs [29], while optimizing its rate-distortion performance via adjusting quantization parameters, which can be seen as an initial trial of Task 2.3. Based on the codec, we also proposed an Adaptive BitRate (ABR) algorithm to enhance the streaming quality by optimizing the codec parameters based on the network conditions. In Fig. 3, we show the sample results without and with our proposed ABR algorithm, demonstrating the merits of our ABR algorithm. These earlier experiences [29, 42] show our expertise in designing scalable dynamic 3DGS codecs.

4 Adaptive Streaming Algorithms

4.1 Proposed Research Task

To stream live deformable avatars and pre-recorded dynamic scenes, we develop an adaptive streaming system for low-latency and high-quality live streaming. In the literature, a few studies [4, 8] combined 3DGS with 3D meshes to represent different objects in bigger virtual worlds. However, they ignored the real-time requirements of streaming applications and did not work for

Table 1: Tentative Project Schedule

Tasks / Quarters	2025-Q3	Q4	2026-Q1	Q2	Q3	Q4	2027-Q1	Q2	Q3	Q4	2028-Q1	Q2
Task 1: Deformable 3DGS Avatars												
1.1: Deformable Avatar Construction												
1.2: User States Estimation												
1.3: Real-time Avatar Updates												
Task 2: Scalable Dynamic 3DGS Codecs												
2.1: Hierarchical Dynamic Representation												
2.2: Dynamic Coding Tools												
2.3: Hybrid Scalable Codecs												
Task 3: Adaptive Streaming Algorithms												
3.1: Distributed Caching Algorithms												
3.2: Resource Prediction Algorithms												
3.3: Transmission Scheduling Algorithms												

immersive communications. On the other hand, several projects constructed immersive communication systems [18, 21, 41], but they did not consider 3DGS representations. This proposal to our best knowledge, is the first work trying to build and optimize an adaptive 3DGS streaming system for immersive communications with the following three research tasks:

- **Task 3.1: Distributed caching algorithms.** To avoid additional delays due to: (i) transmitting deformable avatars and dynamic scenes over congested networks and (ii) rendering virtual worlds on less capable devices, we plan to design caching algorithms for 3D representations. The decisions are made by the user behavior, network bandwidth, and device resources, including free storage space and residual computing power. We consider a continuum set of devices from HMDs to cloud servers, including smartphones and edge servers, for the best possible immersive experience.
- **Task 3.2: Resource prediction algorithms.** We plan to develop an ML-based algorithm to predict available resources, such as networking, computing, and storage, during immersive communication sessions. The predicted resources are optimally allocated to various computing tasks, such as state estimation, avatar deformation, and virtual world rendering for the best immersive experience. They can also be leveraged by the distributed caching policy allowing prefetching larger avatars and scenes.
- **Task 3.3: Transmission scheduling algorithms.** To ensure the timely delivery of 3DGS avatars and scenes, we plan to develop a scheduling algorithm that can quickly adapt to high-frequency changes in network conditions. While being executed on a large number of devices with local knowledge, the scheduling algorithm needs to ensure high and fair immersive experiences across all users, making its design and evaluation both challenging.

4.2 Prior Work

We have done quite a few systems work on real-time streaming with a focus on maximizing user experiences. For example, we developed a cloud VR gaming testbed to evaluate Quality of Services (QoS) metrics and analyze the impact of network variations on HMD users [17]. We also assessed the presence and cyber-sickness scores in VR applications across different 3D representations (including 3DGS) and interaction modes [9]. For point cloud

streaming [16], we proposed a motion prediction algorithm to conceal packet losses during streaming. Furthermore, we proposed the first DASH streaming system for dynamic 3DGS scenes [28], which improves transmission efficiency through layering, tiling, and segmentation while maximizing user experiences. In Fig. 4, we show the sample rendered views from 3DGS streaming systems without and with our optimization techniques. It is not hard to see that, without our optimization techniques, the avatar is highly distorted in both geometry and color, which will turn users away from the system. The resulting design and implementation serve as the basis of our coming explorations in Tasks 3.1–3.3. Our established publication record [9, 16, 17, 28] highlights our capability in developing an adaptive streaming system for 3DGS virtual worlds.

5 Expected Impacts

This proposal aims to address key challenges in streaming dynamic 3DGS virtual worlds over next-generation networks for 6DoF interactions. We divide the project into three major tasks: (i) deformable 3DGS avatars, (ii) scalable dynamic 3DGS codecs, and (iii) adaptive streaming algorithms. These tasks are further divided into nine subtasks, which will be gradually completed. Table. 1 gives the tentative schedule of the proposed subtasks. Our proposed solutions will enable highly immersive, responsive, and scalable communication systems that can be adapted to diverse network conditions, devices, and applications.

Such an immersive communication system can revolutionize various fields. For instance, in telemedicine, immersive remote consultations could enhance healthcare accessibility, particularly in underserved regions. In education, interactive and lifelike virtual worlds could enrich distance learning, making remote instruction more engaging and effective. In remote collaboration, global interactions become possible for professional meetings, creative collaboration, and social engagement. Beyond technical advancements, our work aspires to drive societal impacts by enhancing the accessibility, functionality, and integration of immersive technologies into people's everyday life.

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